Time-Domain Specifications for 2nd Order Underdamped Systems

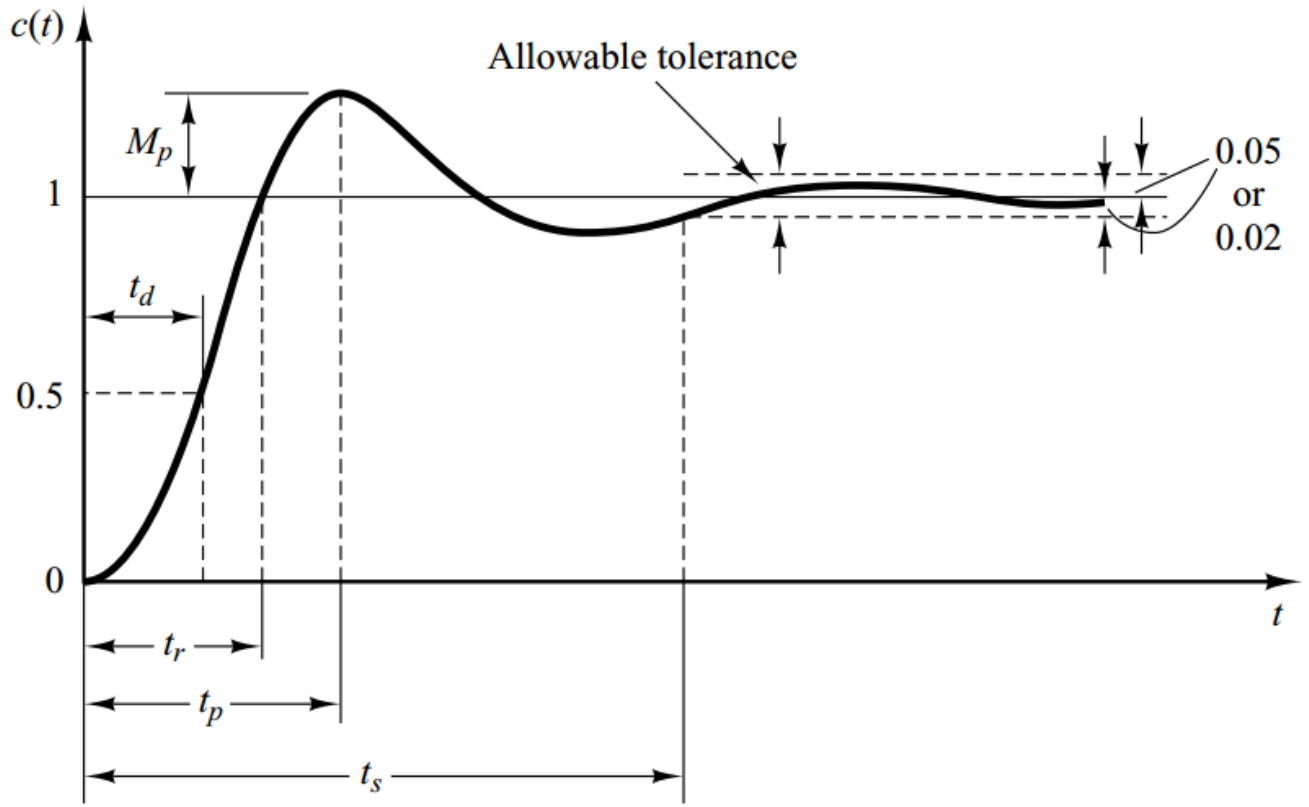
(And Time Response for State Space Equations)

Objectives

- Student should
  - Recognize the typical underdamped second order system time response
  - Describe the following specifications of an underdamped second order system: maximum overshoot, percent overshoot, rise time, settling time, static gain, peak time
  - Find those values from the time plot and from the transfer function
  - Understand the relation between the location of the poles in the s-plane and the time response
  - Be able to find the transfer function from the time response
  - Be able to conclude about higher order systems and approximate them to a lower order system when appropriate.
  - Find the characteristic equation and the eigenvalues of a system describe space state equations
  - Find the time response using state variable equations.

7-3 THE UNIT-STEP RESPONSE AND TIME-DOMAIN SPECIFICATIONS

For linear control systems, the characterization of the transient response is often done by use of the unit-step function $u_s(t)$ as the input.



Underdamped Step Response Specifications

- Maximum Overshoot (M) = Max Value – Final Value
- Percent Overshoot (M_p) = (Max Overshoot/Final Value) x 100
- Peak Time (t_p): time when the maximum occurs
- Rise Time (t_r) has several definitions. (1) The time when the signal rises from the 10% of the final value to the 90% of the final value. (2) The time to reach for the first time the 100% of the final value. We will us the latter.
- Settling Time (t_s): (1) Five percent criterion: the time when the signal gets within a gap of ±5% of the final value. (2) Two percent criterion: the time when the signal gets within a gap of ±2% of the final value.

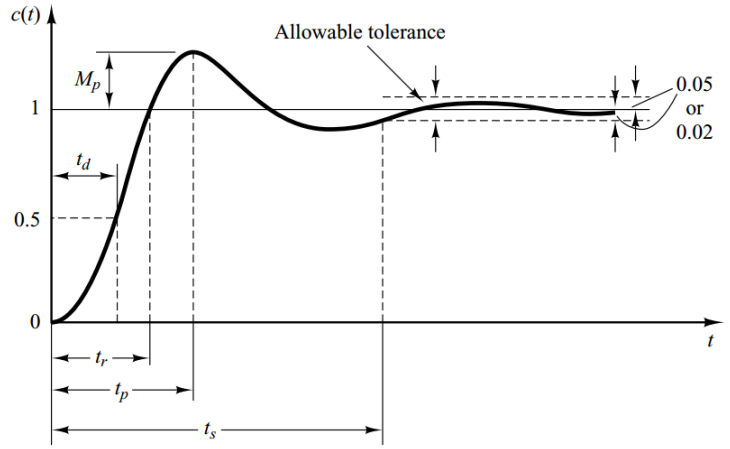
Percent Overshoot

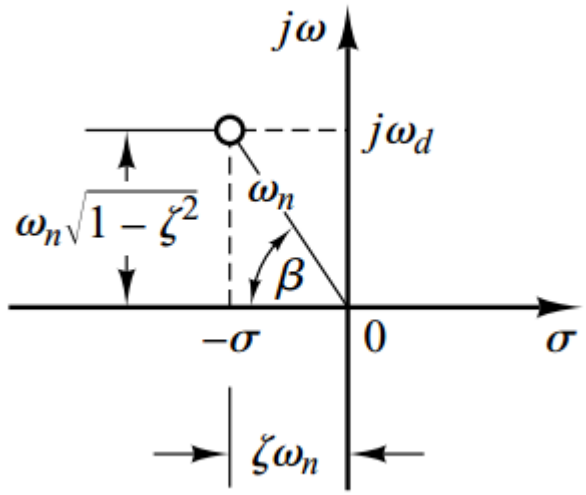
Maximum overshoot occurs at the peak time given by:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

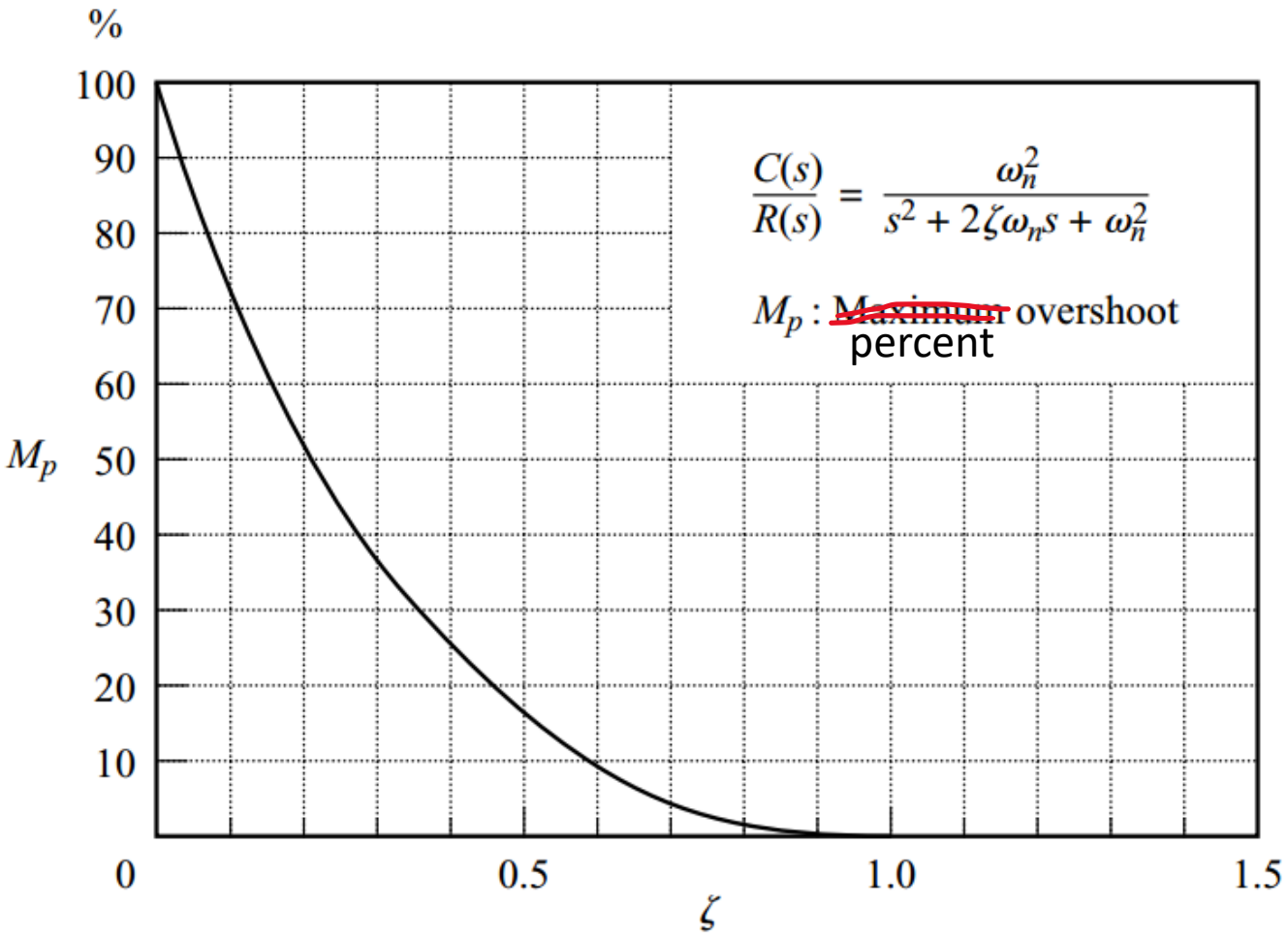
Percent Overshoot is given by:

$$M_p = 100 \times e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$$

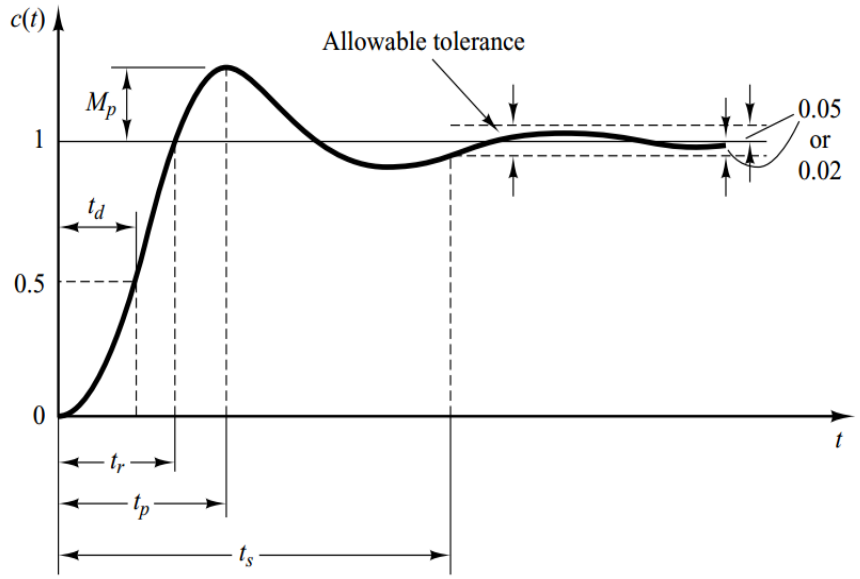




Percent Overshoot Plot



Rise Time



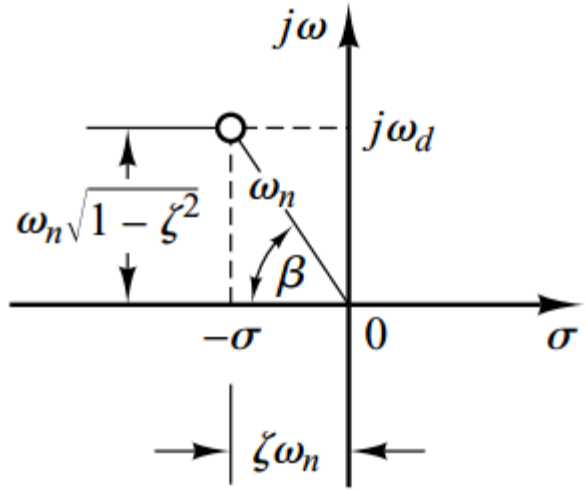
The rise time is the time in which the signal goes from 0 to 100% of the final value. The formula is given by:

$$t_r = \frac{\pi - \beta}{\omega_d}$$

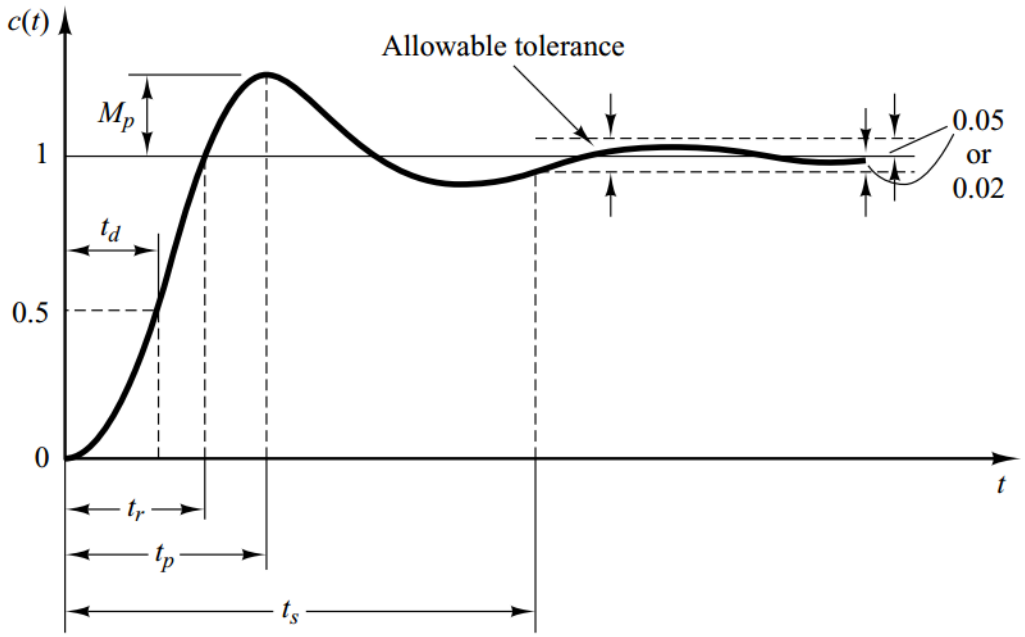
$$\beta = \cos^{-1}\zeta \ \text{(rad)}$$

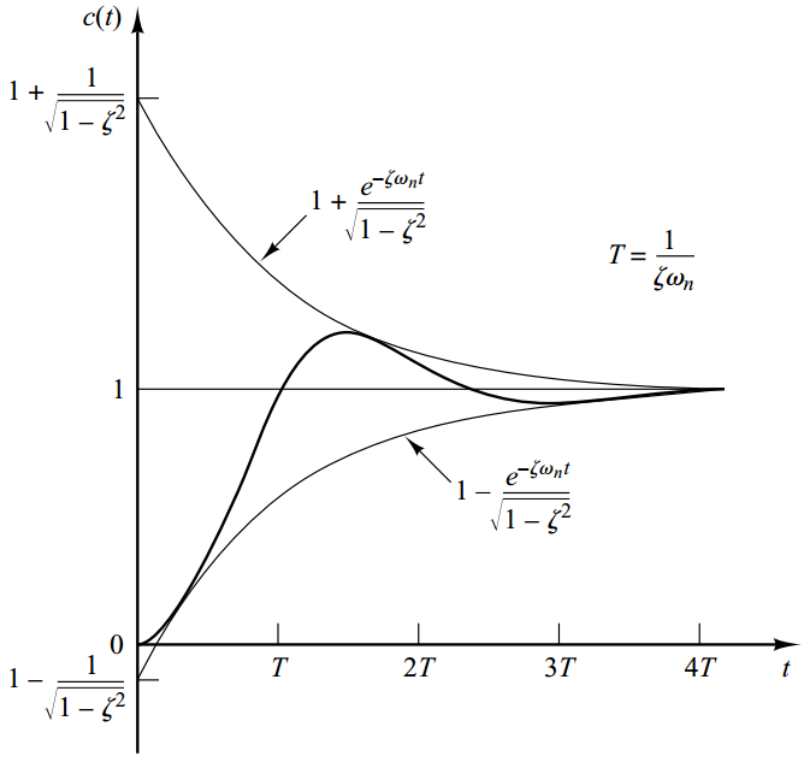
$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

ω_d is the damping frequency (oscillation frequency)



Settling Time (5%)

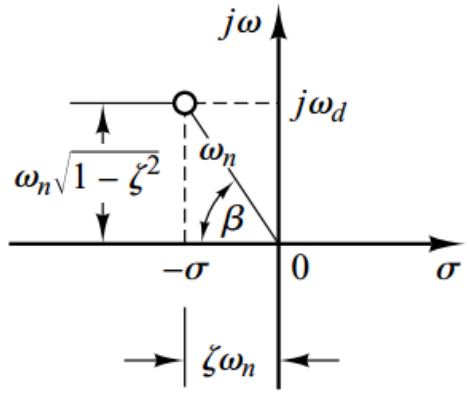




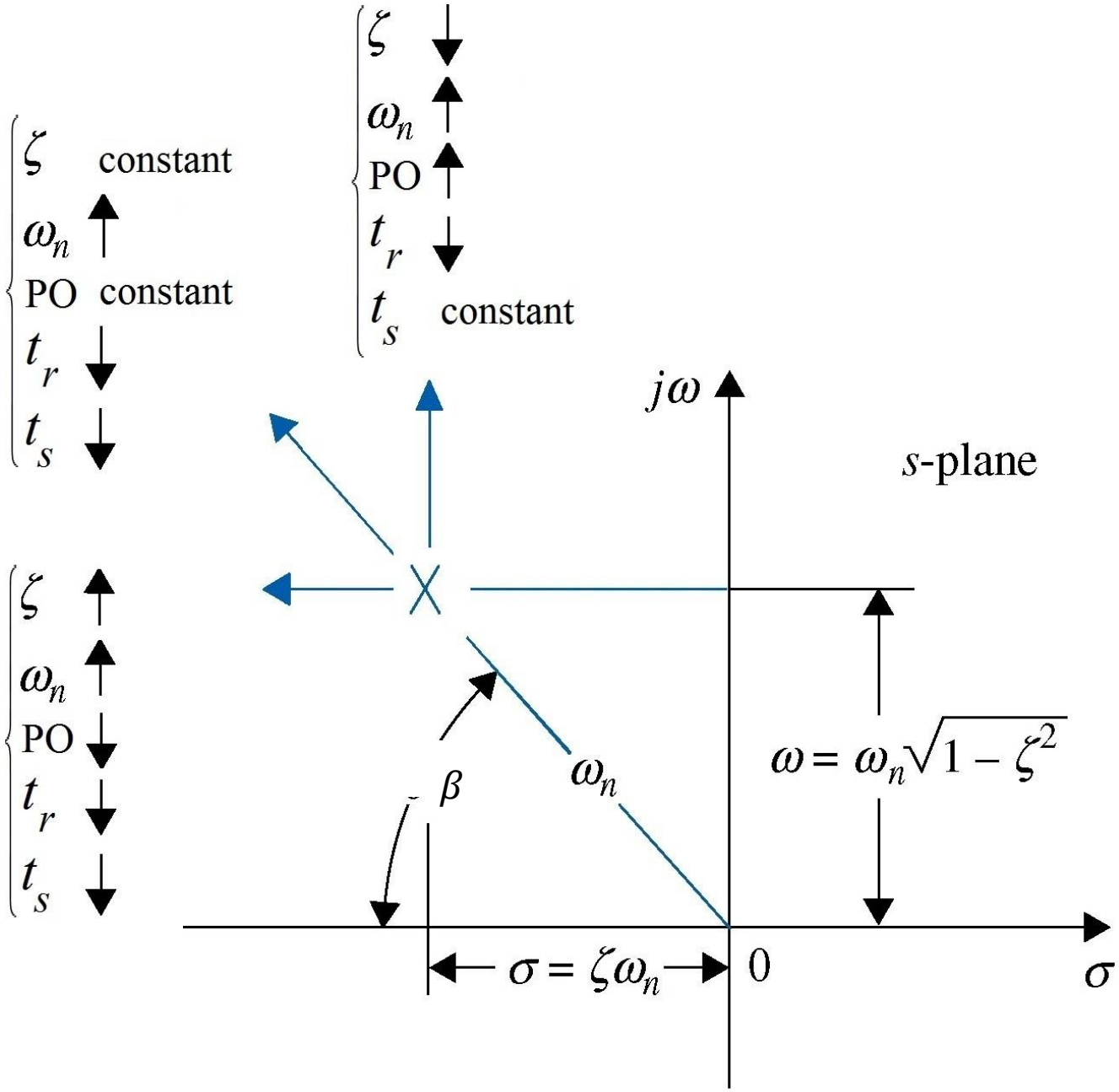
Settling Time Approximation:

$$t_s \lessapprox 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} \qquad \text{(2\% criterion)}$$

$$t_s \approx 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} \qquad \text{(5\% criterion)}$$



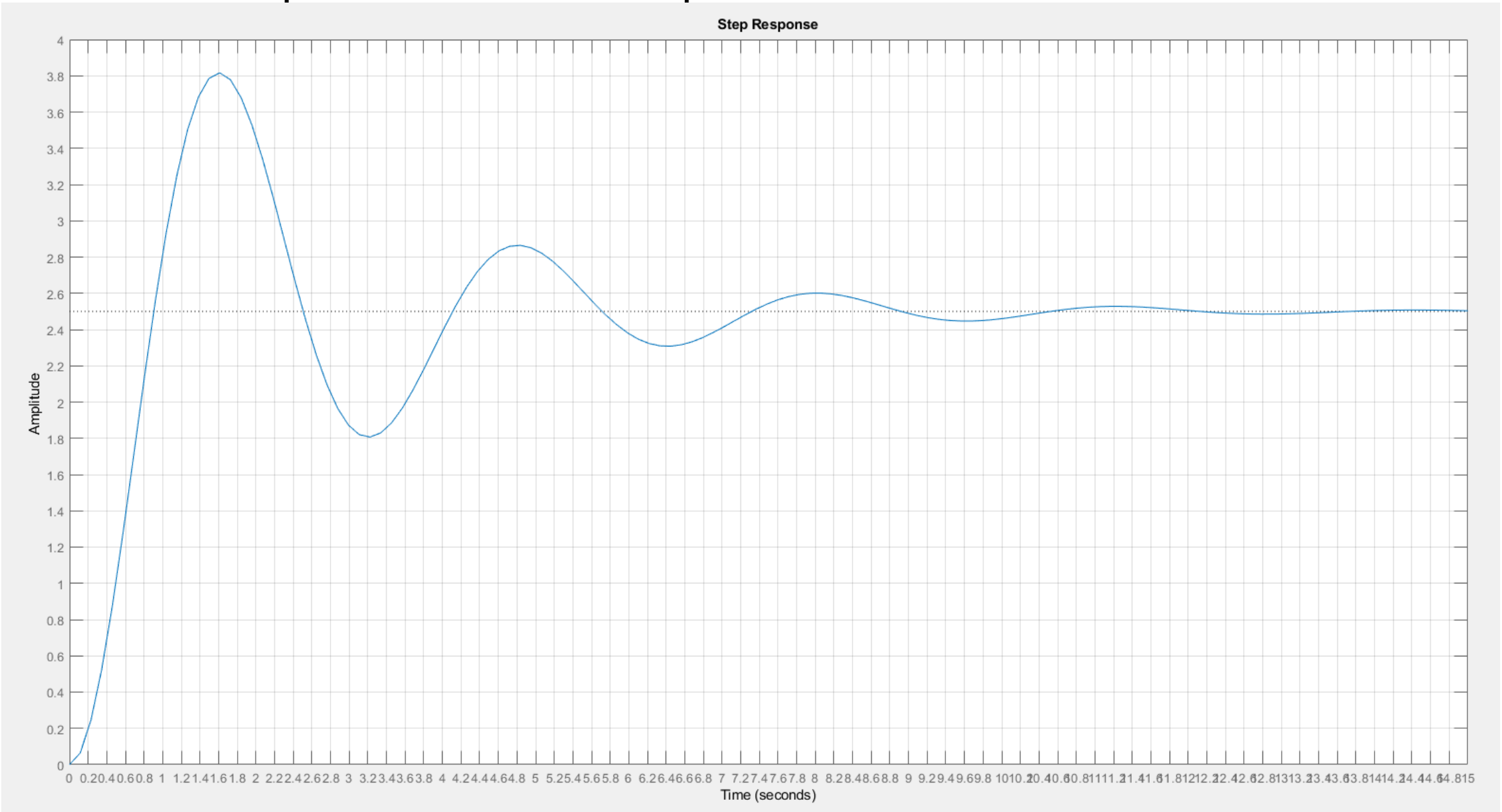
Correlation between pole locations and time response



Example 1

- A second order system is excited with a step of magnitude 1.25. The output response is shown in the figure. Find (a) the percent overshoot, (b) the peak time, (c) the rise time (0 to 100%), (d) the settling time (2% criterion), (e) the final value, (f) the oscillation frequency (rad/s), (g) the damping ratio, (h) the natural frequency (rad/s), (i) the static gain, and (j) the transfer function.

Time response of example 1



Order Reduction

- We have seen that the poles determine the time constants of the exponentials. To be more specific, the real part of the poles.
- A negative real part is associated with a decaying exponential, a positive real with an increasing exponential. This is why we say that poles in the s-RHP are unstable and are undesirable in any control system.
- But even the poles on the jw axis are not good from the control point of view. If a system has a pole at the origin, it is equivalent to have an infinite time constant (never settles). The output will steadily grow. If a system has a pair of conjugate poles on the jw axis, this system is pure oscillatory. It will oscillate and its oscillations won't be damped.

Order Reduction

- An order reduction can be done based on the relative location of the poles. Poles that are far from the origin in relation with another one have a very fast time constant. These poles could be neglected (see example in class), and a reduction of order can be made.
- The other way to reduce the order is by observing the relative position between a zero and a pole. If a zero is very close to a pole (see example in class), this zero could cancel out the pole and so we could reduce the order of the system.
- In a stable system, that is, all its poles are on the left-half plane (LHP) of the s-plane, the dominant poles are those closer to the origin and will dominate the time response.

Control System Specifications

- A control system will be essentially a feedback system (closed loop system)
- The idea is to design a controller that makes the closed loop system have certain dominant poles.
- The dominant poles must produce, for example, good overshot, rise time, settling time, etc.
- We will design controllers using the Root Locus method in order to satisfy requirements such as overshoot, rise time or/and settling time, in order words, in order to assign specific dominant poles.
- Also, the control system must be accurate. It must follow the reference with small error. (See steady state error material in next lectures).